Response to the complete scientific review

This is an internal scientific response, not a message sent to referees. The full review was read before changes. All 19 substantive concerns are tracked below; the complete 59-result and 20-assumption audits remain preserved and feed the dependency audit. P0/P1 correctness, methods and experiments were addressed before rewriting the narrative.

Response matrix

The CSV/JSON matrix includes review location, scientific type, proposed action, actual action, files changed, new result and remaining issue.

ID / priority	Concern	Status
F01 / P0	Coverage tolerance counts a genuine miss	RESOLVED
F03 / P0	No synchronized final submission	RESOLVED
F02 / P1	No-bias ablation changes three components	RESOLVED
F04 / P1	Interval midpoint has catastrophic rare-fallback loss	RESOLVED
F05 / P1	Signal sweep changes supplied norm budget	RESOLVED
F06 / P1	Comparator guarantee type is not applicability	RESOLVED
F07 / P1	Ideal high-dimensional fit versus binary computation	RESOLVED
N01 / P1	Focus the distinct tensor likelihood contribution	RESOLVED
F08 / P2	Two calibration refinements lack direct experiments	RESOLVED
E01 / P2	Pointwise precision and finite-grid interpretation	RESOLVED
D01 / P2	No justified real-data/LLM application	NOT RESOLVED
C01 / P2	Runtime and memory access models differ	RESOLVED
F09 / P3	Scalar losses reused across interval levels without identity	RESOLVED
F10 / P3	Colliding symbols and historical exposition	RESOLVED
T01 / P2	General fixed-cap weak-signal minimax computation	NOT RESOLVED
T02 / P2	Intermediate growing-cap minimax matching	NOT RESOLVED
T03 / P2	Arbitrary loading profiles beyond matched regimes	NOT RESOLVED
T04 / P2	Complete pointwise and TT adaptation theory	NOT RESOLVED
T05 / P2	Unknown directional scale, dependent replicates and selected loading	PARTIALLY RESOLVED

F01: Coverage tolerance counts a genuine miss

Review concern: Coverage tolerance counts a genuine miss. The precise original location is recorded in registry row F01.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Replaced target-scaled tolerance by four endpoint ULPs, retained strict counts, reevaluated all saved interval observations and regenerated dependent outputs.

Technical resolution / new evidence: Exactly two 90% rows for the same S03_007 observation change from 134/150 to 133/150; 606 rescaling checks pass.

Manuscript and artifact changes: numerics.py, recompute_saved.py, saved_corrections.json.

Remaining issue: This is numerical evaluation rounding, not formal interval arithmetic.

Status: RESOLVED.

F03: No synchronized final submission

Review concern: No synchronized final submission. The precise original location is recorded in registry row F03.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Created one current main paper, complete technical supplement, empirical supplement and reading order; integrated final rates and open boundaries.

Technical resolution / new evidence: All 59 reviewed results plus R1 have an explicit proof-status ledger and complete supplied proofs; the final rate figure replaces the historical gap headline.

Manuscript and artifact changes: main.tex, technical_supplement.tex, empirical_report.md, README.md.

Remaining issue: Final venue formatting and targeted citation/priority check remain.

Status: RESOLVED.

F02: No-bias ablation changes three components

Review concern: No-bias ablation changes three components. The precise original location is recorded in registry row F02.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Executed paired isolated bias, quantile and intersection ablations; derived pure fallback removal exactly from every saved tangent candidate. Relabeled the old multicomponent diagnostic.

Technical resolution / new evidence: Bias-only removal changes nothing over the first eight V01 levels because fallback is selected. Pure fallback removal retains intersections; the combined failure cannot identify bias alone.

Manuscript and artifact changes: run_revision.py, analyze_revision.py, revision_summary.csv, 03_controlled_ablations.pdf.

Remaining issue: These cases do not establish that every correction is necessary at every mean.

Status: RESOLVED.

F04: Interval midpoint has catastrophic rare-fallback loss

Review concern: Interval midpoint has catastrophic rare-fallback loss. The precise original location is recorded in registry row F04.

Assessment: The empirical concern is valid. The prior P1 theorem asserted expected length, not midpoint MSE, so no false P1 theorem is being repaired. The estimator output nevertheless required a substantive correction.

Action: Introduced a separate public scalar output with a complete uniform risk proof; reevaluated all affected original observations and retained old losses.

Technical resolution / new evidence: R1 risk is at most C_alpha $\min(h^2,\tau^2)$. S03_002 MSE changes from 12,084,661.379 to 0.147776 on the same draws; intervals are unchanged.

Manuscript and artifact changes: scalar_companion.md, methods.py, numerics.py, scalar_risk_saved.csv.

Remaining issue: P1 pointwise minimax MSE remains unproved; R1 is only a noise-scale bound.

Status: RESOLVED.

F05: Signal sweep changes supplied norm budget

Review concern: Signal sweep changes supplied norm budget. The precise original location is recorded in registry row F05.

Assessment: The criticism is valid as a design confounding issue. A supplied norm bound chosen for each prescribed simulation class is not data leakage; it simply changes the question being measured.

Action: Executed a fixed-M sweep with a paired norm-comparable-M branch, holding the observations and pilot fixed between branches.

Technical resolution / new evidence: Ten amplitudes and 10,000 new observations distinguish signal strength from a changing supplied honesty class.

Manuscript and artifact changes: revision_cases.json, run_revision.py, 02_fixed_budget.pdf.

Remaining issue: Both branches still assume the supplied norm/cap bounds are valid.

Status: RESOLVED.

F06: Comparator guarantee type is not applicability

Review concern: Comparator guarantee type is not applicability. The precise original location is recorded in registry row F06.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Mapped the XCH implementation to tensor-PCA Sections 4.2–4.4 and added case-level applicability separately from guarantee type.

Technical resolution / new evidence: Zero/rank mismatch explicitly outside the specified regular model; other initialization, alignment and remainder constants not certified. Smaller strong-signal comparator lengths retained.

Manuscript and artifact changes: novelty_and_baselines.md, recompute_saved.py, analyze_revision.py, empirical_results.tex.

Remaining issue: This is a local numerical implementation, not certification of every finite-case regularity constant.

Status: RESOLVED.

F07: Ideal high-dimensional fit versus binary computation

Review concern: Ideal high-dimensional fit versus binary computation. The precise original location is recorded in registry row F07.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Separated ideal residual optimization, executable conservative methods and binary float64 search; fixed loading, direction and tolerance across 600 new solves.

Technical resolution / new evidence: 7/600 endpoint certificates, 593 unresolved; all witness gaps and observations retained. The C1 objective-error contract remains explicit.

Manuscript and artifact changes: main.tex, run_revision.py, solver_trials.csv, 06_fixed_solver.pdf.

Remaining issue: General high-dimensional minimax computation remains open under T01.

Status: RESOLVED.

N01: Focus the distinct tensor likelihood contribution

Review concern: Focus the distinct tensor likelihood contribution. The precise original location is recorded in registry row N01.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Rebuilt the contribution around full-likelihood tensor geometry, the cubic harmonic mixture and all-budget ordered-coordinate bound; explicitly credited classical components.

Technical resolution / new evidence: Distinguishes changes in observation experiment, honesty class and estimator restriction. No claim of generic new nonlinear inference or firstness.

Manuscript and artifact changes: novelty_and_baselines.md, introduction_outline.md, main.tex.

Remaining issue: External priority confirmation is a submission risk; this is a focused mathematical positioning audit.

Status: RESOLVED.

F08: Two calibration refinements lack direct experiments

Review concern: Two calibration refinements lack direct experiments. The precise original location is recorded in registry row F08.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Executed the actual row-corruption median and independent Gaussian residual-scale refinements, including assumption violations.

Technical resolution / new evidence: 19,000 row/noise datasets and 54,000 variance datasets. The severe cap violation fails; independent scale calibration is conservative and naive plug-in fails at small n.

Manuscript and artifact changes: run_revision.py, calibration.csv, 05_calibration.pdf.

Remaining issue: No general contamination-minimax or arbitrary-unknown-noise claim.

Status: RESOLVED.

E01: Pointwise precision and finite-grid interpretation

Review concern: Pointwise precision and finite-grid interpretation. The precise original location is recorded in registry row E01.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Added four p=32/64 cases with 1,000 independent trials each and a 1,000-trial transition grid; report normalized lengths and Wilson uncertainty.

Technical resolution / new evidence: Large-case 95% cap coverage 0.965–0.978; rate-normalized length approximately 7.20–8.01. All rare losses retained.

Manuscript and artifact changes: revision_cases.json, revision_summary.csv, 04_scalar_and_pointwise.pdf.

Remaining issue: Finite grid does not prove a limiting exponent, minimal SNR or general pointwise geometry.

Status: RESOLVED.

D01: No justified real-data/LLM application

Review concern: No justified real-data/LLM application. The precise original location is recorded in registry row D01.

Assessment: No invalid real-data result existed to repair. Adding an arbitrary dataset would not establish the required sampling model or prescribed target. A real-data extension remains unavailable rather than being marked scientifically resolved.

Action: Audited the absence of a justified repeated-tensor dataset and contrast; kept a theory/simulation scope and stated the required independent sampling unit and calibration.

Technical resolution / new evidence: No unsupported application was added. A text clarification was requested; no dataset/model specification was received during revision.

Manuscript and artifact changes: README.md, main.tex, empirical_report.md.

Remaining issue: A defensible real-data analysis is not yet available. It is not needed for the focused theory claim but remains an open broader project request.

Status: NOT RESOLVED.

C01: Runtime and memory access models differ

Review concern: Runtime and memory access models differ. The precise original location is recorded in registry row C01.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Measured the revised public interval/scalar API with warmup and ten calls per dimension; separated input, coordinate access and traced allocation.

Technical resolution / new evidence: Fresh p=8/16/32/64 timings, explicit p^3 entries and 8p^3 input bytes; coordinate access 8p bytes. Economy-SVD cost stated.

Manuscript and artifact changes: check_methods.py, runtime_revision.json, empirical_report.md.

Remaining issue: Trace excludes input and possibly native workspace and is not RSS; no sublinear full-tensor computation claim.

Status: RESOLVED.

F09: Scalar losses reused across interval levels without identity

Review concern: Scalar losses reused across interval levels without identity. The precise original location is recorded in registry row F09.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Added estimator_id and estimator_alpha; distinguish 95% scalar outputs from reused multi-level interval rows and retain midpoint diagnostics separately.

Technical resolution / new evidence: All scalar-producing interval summaries identify the estimator; confidence levels are not counted as independent data.

Manuscript and artifact changes: common.py, analyze_revision.py, revision_summary.csv.

Remaining issue: Event-only records have no scalar estimator and are labeled accordingly in the revision guide.

Status: RESOLVED.

F10: Colliding symbols and historical exposition

Review concern: Colliding symbols and historical exposition. The precise original location is recorded in registry row F10.

Assessment: The concern is valid. Its scope must be distinguished from any stronger unproved interpretation.

Action: Created a notation dictionary, separated ambient dimension, polynomial degree, envelope coefficients and approximation/pilot quantities, and reorganized historical bounds.

Technical resolution / new evidence: A focused main argument and complete proof supplement replace the chronological chain of drafts; active reference labels resolve.

Manuscript and artifact changes: notation.md, prepare_theory.py, main.tex.

Remaining issue: Locally introduced proof aliases are documented; publication style can receive a final copy edit.

Status: RESOLVED.

T01: General fixed-cap weak-signal minimax computation

Review concern: General fixed-cap weak-signal minimax computation. The precise original location is recorded in registry row T01.

Assessment: The concern identifies an unresolved research extension. The focused manuscript can state the existing results without claiming this extension.

Action: Re-examined objective certificates, truncated-scan tradeoffs and restricted computational obstructions; executed fixed-geometry solver diagnostics.

Technical resolution / new evidence: The numerical and statistical contracts are concrete; no general polynomial minimax algorithm was obtained.

Manuscript and artifact changes: revision_audit.md, main.tex, solver_summary.csv.

Remaining issue: Need a general algorithm with deterministic global error control or a correctly scoped computational lower bound.

Status: NOT RESOLVED.

T02: Intermediate growing-cap minimax matching

Review concern: Intermediate growing-cap minimax matching. The precise original location is recorded in registry row T02.

Assessment: The concern identifies an unresolved research extension. The focused manuscript can state the existing results without claiming this extension.

Action: Compared explicit all-r sparse-mixture lower and structural/direct upper envelopes, retaining rank dependence and the preserved intermediate-cap grid.

Technical resolution / new evidence: Large-cap matching is preserved; extrapolating fixed-cap constants does not match intermediate caps.

Manuscript and artifact changes: revision_audit.md, main.tex, 10_intermediate_caps.pdf.

Remaining issue: Sharp matching rates for $1 \ll r \ll \sqrt{p}$ remain open.

Status: NOT RESOLVED.

T03: Arbitrary loading profiles beyond matched regimes

Review concern: Arbitrary loading profiles beyond matched regimes. The precise original location is recorded in registry row T03.

Assessment: The concern identifies an unresolved research extension. The focused manuscript can state the existing results without claiming this extension.

Action: Re-examined dyadic upper and flat-block/largest-atom lower bounds and the deterioration as decay tends to zero.

Technical resolution / new evidence: Flat, small-budget arbitrary and fixed-decay regimes are precisely separated.

Manuscript and artifact changes: revision_audit.md, main.tex.

Remaining issue: General slowly varying, multiscale and nonorthogonal profiles at larger budgets still lack matching theory/computation.

Status: NOT RESOLVED.

T04: Complete pointwise and TT adaptation theory

Review concern: Complete pointwise and TT adaptation theory. The precise original location is recorded in registry row T04.

Assessment: The concern identifies an unresolved research extension. The focused manuscript can state the existing results without claiming this extension.

Action: Checked the role of saturation, norm slack, pilot signal and growing-cap obstructions; added R1 without claiming pointwise minimax risk.

Technical resolution / new evidence: P1 length and G2 impossibility remain valid within their original scopes; scalar rare-event loss is repaired.

Manuscript and artifact changes: revision_audit.md, scalar_companion.md, main.tex.

Remaining issue: Complete pointwise length/risk, optimal SNR and TT-specific adaptation remain open.

Status: NOT RESOLVED.

T05: Unknown directional scale, dependent replicates and selected loading

Review concern: Unknown directional scale, dependent replicates and selected loading. The precise original location is recorded in registry row T05.

Assessment: The concern identifies an unresolved research extension. The focused manuscript can state the existing results without claiming this extension.

Action: Implemented and tested the available independent Gaussian scale refinement and finite-variance row bound; retained explicit dependence/selection restrictions.

Technical resolution / new evidence: Finite-sample calibration directly tested in its real experiment; naive plug-in counterexample retained.

Manuscript and artifact changes: revision_audit.md, calibration.csv, main.tex.

Remaining issue: Unknown arbitrary directional scale, dependent replicates, selected loading and estimated allowances need additional theory.

Status: PARTIALLY RESOLVED.